

Grounded Q&A over Enterprise Documents

Avathon AI/ML Hiring Challenge — Track D (RAG / LLM Knowledge Systems) × Scenario S2 · Raushan Kumar · github.com/Raushan0502/avathon-rag-s2

The brief requires a written justification addressing four things explicitly. This is that justification. Every figure quoted here is regenerable from the repository — `python scripts/report_metrics.py` prints all of them from the artifacts in `results/`.

System, in one line: 100 real SEC EDGAR / NIST / email documents across three formats → 9,409 token-budgeted chunks → `bge-small` embeddings → FAISS `IndexFlatIP` → dense retrieval → cited, refusal-capable generation, evaluated on 45 held-out questions.

1. Chunking strategy, and what is lost at chunk boundaries. Token-budgeted, structure-preserving chunking with contextual enrichment: split on the document's own headings first, then pack blocks to a **360-token budget with 60-token overlap**, never splitting a table row, prepending a document › section header to each chunk's embed text. Rejected: fixed-size character windows (cut mid-sentence and mid-table, and lose the header context needed to tell twenty different "Item 7"s apart); one-chunk-per-section (10-K Item 7 far exceeds any context window); embedding-based semantic splitting (cost scales with corpus, and the gain is unproven on filings that already carry explicit headings). *What is genuinely lost:* (a) **long-range coreference** — a chunk opening "These amounts exclude..." loses its antecedent; overlap and the section header mitigate but do not remove this; (b) **cross-section reasoning** — a question needing Item 7 and Item 8 requires two chunks retrieved together, which nothing in the chunker guarantees. That is the dominant residual error mode. The budget is enforced, not assumed: an earlier build silently truncated 80 chunks past the encoder's 512-token limit; an explicit oversize split plus a 60-token header reserve brought that to **0**.

2. Embedding model choice. BAAI/`bge-small-en-v1.5` (384-dim), chosen by measurement against `bge-large-en-v1.5` (1024-dim) on the same eval set:

Model	Dim	MRR	Recall@k	P@k	Attainment	Query ms
bge-small	384	0.948	0.978	0.554	75%	64
bge-large	1024	0.940	0.978	0.604	82%	531

The result is a genuine trade, not a clean win. `bge-large` is *better* at filling the top-k with relevant chunks (precision 0.604 vs 0.554, attainment 82% vs 75%), *marginally worse* at placing the single best chunk first (MRR −0.008, which on 45 questions is roughly two chunks shifting one rank — noise), and identical on recall. So capacity does buy something here; it buys precision, not ranking.

It ships anyway as **bge-small, on cost:** 1024 dims cost **8.3× more per query** (531 ms vs 64 ms — a permanent runtime tax, not a one-off) and ~1.3 s per chunk to embed, which extrapolates to roughly **3.3 hours** to index the full 9,409-chunk corpus on this CPU against minutes for `bge-small`. On a GPU-backed deployment where query latency is amortised, **bge-large would be the defensible pick** — the precision gain is real. On a CPU-only 48-hour budget it is not.

Caveat, stated because it bounds the claim: this comparison ran on a **200-chunk subset** (every gold chunk plus deterministic distractors), not the full corpus, because a full-corpus `bge-large` pass ran 10.7 hours and was abandoned. The easier candidate pool inflates both models (MRR 0.948 here vs 0.781 full-corpus) and may compress the gap between them. It is sound as a *relative* comparison and should not be read as absolute performance.

Both models are used asymmetrically (query-instruction prefix on queries, none on passages) per the model card.

3. How fusion compares to a single retriever — quantitatively.

Mode	P@k	Recall@k	MRR	Attainment
dense	0.378	0.889	0.781	50.2%
bm25	0.208	0.733	0.555	27.6%
hybrid (RRF, k=60)	0.300	0.867	0.735	39.9%

Hybrid did not beat dense, which is the opposite of the expected result and of what an earlier draft of this project asserted before the numbers were in. The cause is **corpus-wide IDF dilution**: as the corpus grew 6 → 100 documents, the terms that discriminate in filings ("revenue", "Item 7", "Company") appear everywhere, BM25's IDF flattens, and fusing a weaker ranker into a stronger one drags the stronger one down. RRF is kept behind a flag because it is the right default on a lexically diverse corpus, but **dense is the shipped mode here** because that is what the measurement supports.

Precision is also structurally capped — a third of questions have single-chunk gold sections, so $P@5$ cannot exceed 0.2 for them — which is why **attainment** ($P@k \div \text{ceiling } 0.752$) and **MRR** are the honest headline metrics, and why questions are tiered by $k = 1/5/10/20$.

4. Hallucination prevention and detection. Three layers, because prevention alone is not verifiable. *Prevention*: the prompt constrains the model to retrieved context, requires a bracketed citation per claim, and explicitly permits refusal. *Detection*: every answer is parsed for citation markers and annotated `cited` / `refused` / `UNGROUNDED` — **41 cited, 4 refused, 0 UNGROUNDED** on 45 questions, where the refusals are correct behaviour because retrieval genuinely missed ($\text{hit}@k = 40/45$). *Correctness, measured separately*: **cited $\neq$ correct** — grounding says an answer pointed at context, not that it is right. Accuracy is therefore scored independently by a deterministic lexical key-fact scorer and by an LLM judge, because they disagree informatively, and disagreement is where a human should look:

Answer accuracy (41 answered, 4 refused)	Value
Mean lexical key-fact recall	0.567
Lexical accuracy ($\geq 60\%$ of reference key facts)	48.8%
LLM judge — CORRECT	58.5% (24/41)
LLM judge — CORRECT + PARTIAL	85.4% (35/41)
LLM judge — WRONG	6

The two scorers disagree by ~10 points because the lexical one punishes paraphrase — "repeatedly slashing prices" against a reference of "aggressively cut prices" is correct and scores badly. Reporting both, rather than the flattering one, is the point.

Known limitations, stated rather than hidden. (i) The **6 WRONG answers are all grounded** — they cite real retrieved context and are still incorrect, which is why faithfulness alone is an insufficient safety metric and why the judge exists. (ii) The $k=1$ tier scores **$P@k$ 0.000 with MRR 0.278** — golds sit at ranks 2–3, and for $k=1$ the rank-1 result is a *different* Apple RSU exhibit whose clause text is verbatim-identical to the gold, so no retriever could separate them; that is a corpus property, not a ranking failure. (iii) Cross-section questions remain unsolved (§1). (iv) The single highest-value next step is **not** a bigger embedding model but a **cross-encoder reranker** — recall is already 0.889 while precision is 0.378, meaning the right chunk is usually retrieved but not ranked first, which is exactly what reranking fixes.